sends signals from one part to another, and see if the pathways change based on efficiency (as they would if naturally selected).

This is being tested by studying human brains, by studying human behaviour, and by building AI based on the learnings of this field to see if AI can evolve as well. Thus far, AI has not been able to "evolve" because mutations that we program in don't always select for something better, whereas the proponents of this new field believe that evolution will essentially build better and better machines (both flesh-based or silicon-based!).

The work being done in this field is looking to use their technique in AI robots, to teach them, to say, walk over different types of materials, or get up when they fall, much in the way a baby or toddler "learns" how to do these things. We humans go from very unsteady little children to being more sure on our feet (well most of us), and in theory, this field of study will not only help us understand our own consciousness, but build better AI.

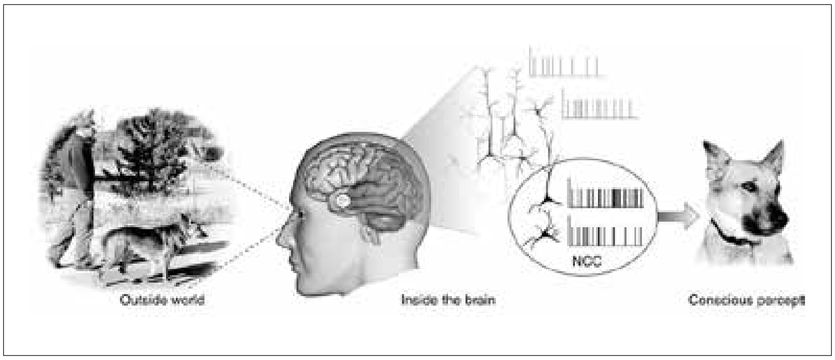
THE PHYSICS OF CONSCIOUSNESS

In 2016, Edward Witten, widely regarded as one of the smartest physicists alive and major proponent of String theory (and M theory) – the proposed theory of everything, revealed in an interview that the concept of consciousness might always remain a mystery. Here was a physicist who deals with imagining 21 dimensions on a daily basis, who thought that an explanation of consciousness might always elude scientists. A theory of consciousness means something totally different to a physicist than it does a biologist – although both seem to be flummoxed by it.

We can at best come up with a fuzzy explanation for what it is – consciousness is the ability of the mind to be aware of what is happening around us. While this definition itself raises a multitude of questions, let us assume for the sake of this piece that this definition suffices. Two distinct ways of tackling this major research problem have come up – the physicists' way and the neurobiologists' way. Physicists like to predict the way matter around us *behaves*. They encapsulate this behavior by a set of rules built around the language of mathematics. Can consciousness be described in the same language?

Neurobiologists, as we've shown above want to pinpoint the region in the brain that governs consciousness. Even if biologists succeed in the quest of mapping each known human function to a particular region in the brain, there is the hard pressing question of how gray and white matter combine to enable us with a highly subjective and perspective view of the world around us.

Of course, it may also be that consciousness might naturally arise from a feature of nature that is so fundamental that it remains hidden and hence unknown to us. This seemingly insurmountable hurdle, however, hasn't stopped physicists from trying to describe consciousness. Let's look at some of the best efforts and theories, none of which have been validated, and in fact, many of which are still hotly debated.



A representative image to demonstrate common understanding of consciousness

QUANTUM MECHANICS

Quantum Mechanics is usually the first stop for all things physics these days, because it's the golden feather in the hat of twentieth-century theoretical physics. Quantum mechanics has long been associated with being the 'weird' theory that explains 'weird' behavior at the atomic scale. It only fits that a quantum mechanical theory of consciousness would be floated at some time or other. For starters, quantum mechanics postulates that rather than any property having an absolute value, it is smeared out as probability governed by a wave function. Any number that a measurement device throws up is because on making this measurement the wave function 'collapses' to give us the one specific value of the property. Essentially according to the dictat of quantum mechanics, objects (at the quantum level of course) are allowed to coexist in a superposition of states. This counterintuitive effect is made use of to create quantum computers out of units called qubits that act as computing devices. According to one of the most controversial theories in modern time, qubit like units might be responsible for consciousness – and quantum effects might be responsible for driving cognition.

Roger Penrose, a well known British physicist, hypothesized way back in the 1980's that there must be a deep link between quantum mechanics and consciousness. At the crux of his theory lies the ability of individual units in the brain to behave like quantum qubits and exist in a superposition of multiple states – each also having the ability to 'collapse' to one single state. A bunch of such qubit like units in one state acting together as a collective phenomenon could lead to what physicist's like to call quantum coherence. In collaboration with physician Stuart Hameroff, the duo even suggested a candidate for qubits in the brain – microtubules. These are protein like structures which are tubular and help secure synaptic connections in neurons. Often called Orch-OR (Orchestrated Objective Reduction), the theory makes use of Penrose's other debated theories on quantum gravity to, in simpler terms, claim the following: